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Lithium niobate piezo-optomechanical crystals

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Demonstrating a device that efficiently connects light, motion, and microwaves is an outstanding challenge in classical and quantum photonics. We make significant progress in this direction by demonstrating a photonic crystal resonator on thin-film lithium niobate (LN) that simultaneously supports high-Q optical and mechanical modes, and where the mechanical modes are coupled piezoelectrically to microwaves. For optomechanical coupling, we leverage the photoelastic effect in LN by optimizing the device parameters to realize coupling rates $g_0/2\pi \approx 120$ kHz. An optomechanical cooperativity $C > 1$ is achieved leading to phonon lasing. Electrodes on the nanoresonator piezoelectrically drive mechanical waves on the beam that are then read out optically allowing direct observation of the phononic bandgap. Quantum coupling efficiency of $\eta \approx 10^{-8}$ from the input microwave port to the localized mechanical resonance is measured. Improvements of the microwave circuit and electrode geometry can increase this efficiency and bring integrated ultra-low-power modulators and quantum microwave-to-optical converters closer to reality. © 2019 Optical Society of America under the terms of the OSA Open Access Publishing Agreement

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1. INTRODUCTION

Optomechanical crystals (OMCs) provide a powerful platform for engineering interactions between photons and phonons. Colocalizing optical and mechanical modes allows light to control and read out mechanical motion, leading to a large variety of quantum optomechanical experiments [1–5] and enabling routing and transduction of classical and quantum signals between microwave frequency phonons and optical photons [6–10]. Silicon optomechanical crystals have strong coupling rates but lack intrinsic piezoelectricity [11]. Capacitive and electrostrictive forces in thin-film silicon have been used to drive mechanical resonances [12–14], but a viable way of efficiently coupling to the high-frequency wavelength-scale mechanical modes of an optomechanical crystal resonator is yet to be demonstrated. Thus approaches to coupling OMCs to microwave photons and superconducting qubits have focused on piezoelectric materials such as aluminum nitride (AlN) [15–18], gallium arsenide (GaAs) [19–23], and gallium phosphide (GaP) [24,25]. However, no platform to date has succeeded in simultaneously demonstrating high optical and mechanical quality factors, large optomechanical coupling, and good piezoelectric coupling efficiency. Recent work on nanopatterned lithium niobate (LN) has addressed several of these weaknesses, by showing that high mechanical Q and efficient piezoelectric transduction of localized modes are possible [26,27], while excellent optical Qs have been demonstrated in photonic crystals [28,29] and microring structures [30]. In this work we demonstrate an LN device with sufficiently large optical and mechanical Qs and optomechanical coupling to realize efficient optical transduction of microwave frequency phonons. Considering that efficient microwave transduction has already been realized in the same material system with phononic crystal devices [26], our result solves a key remaining technical challenge required for an LN quantum microwave-to-optical transducer.

Here we demonstrate one-dimensional (1D) nanobeam OMCs on LN with optical quality factors above 300,000 at a wavelength of $\lambda \approx 1550$ nm, and a mechanical mode frequency close to 2 GHz possessing quality factors as high as 17,000 at 4 kelvin. The high-frequency mechanical mode and narrow optical linewidth allow our system to operate in the resolved-sideband regime, where the mechanical frequency greatly exceeds the cavity decay rate. An important parameter of the system is the zero-point coupling rate g_0 that characterizes the optical frequency jitter induced by the zero-point motion of the cavity. We make calibrated measurements of optomechanical backaction in addition to measurements of electromagnetically induced transparency (EIT) and absorption (EIA) with different optical pump powers and detunings to extract a zero-point coupling rate of $g_0/2\pi \approx 120$ kHz. The wide bandgap (~3.7 eV) of LN allows us to pump $n_c \sim 1.5 \times 10^5$ intracavity photons without significant degradation of the optical linewidth. This enables us to realize sufficiently large cooperativity $C \equiv 4g_0^2 n_c/\kappa\gamma_i$ to observe mechanical lasing under the action of a blue-detuned optical pump, where κ and γ_i are the optical and intrinsic mechanical linewidths, respectively. Phonon lasing is an important signature of optomechanical cooperativity $C > 1$, which occurs when the optomechanical backaction exceeds the intrinsic mechanical dissipation in the system, a requirement for realizing efficient phonon–photon conversion [31–35]. Our system performs favorably

when compared to other piezo-optomechanical approaches. In particular, OMCs on GaAs show $g_0/2\pi$ on the order of 1 MHz [20] but have not operated in the sideband-resolved regime due to low optical $Q \sim 40,000$. Moreover, $g_0/2\pi \sim 100$ kHz has been achieved on AlN [18], but our LN device possesses somewhat higher optical and mechanical quality factors. Finally we note that all three platforms, AlN [36], GaAs [37], and LN [26,38], have demonstrated compatibility with superconducting qubit technology and form promising platforms for quantum transduction.

The piezoelectric property of LN enables us to electrically drive the localized mechanical mode with an oscillating microwave electric field using fabricated on-chip electrodes. The electrically generated phonons are detected optomechanically allowing us to measure the piezo-optomechanical response of the OMC. More generally, electrical driving allows us to probe the full mechanical spectrum of the device with far greater sensitivity than is achieved by measuring thermal Brownian motion, as it allows us to drive the modes to a larger amplitude than is possible thermally. This leads to a remarkably clear signature of the mechanical bandgap of the nanobeam mirror unit cell that is visible from the piezo-optomechanical response. Furthermore, we analyze the piezo-optomechanical coupling by calibrating the coherent phonon number from the electrical drive against the known thermal phonon occupancy to infer the effective microwave-mechanics coupling rate $g_\mu/2\pi \sim 2$ kHz that characterizes the rate at which microwave frequency phonons would interact with a microwave frequency resonator (assuming a 50 Ω resonator impedance). Compared to the mechanical losses, this coupling is about 2 orders of magnitude away from achieving sufficiently large cooperativity to enable coherent microwave-to-optical conversion.

We have demonstrated that the LN on a silicon platform can be engineered to support strong optomechanical coupling between telecom photons and gigahertz phonons. This further demonstrates that LN is a material system with enormous potential for quantum electro-optic and optomechanical transduction [17,21,26,39–42]. In addition to piezo-optomechanical transduction, the electro-optic and nonlinear optical properties of LN enable active integrated optical elements and enhanced optical parametric processes [30,40,41,43,44].

2. DESIGN

We begin with the design and simulation of the 1D OMC on LN. Owing to imperfections in the LN nanofabrication process, the resulting inside and outside sidewall angles are approximately $\theta_{\text{in}} = 22°$ and $\theta_{\text{out}} = 11°$ from the vertical direction. As a result, the nanobeam no longer possesses the reflection symmetry about the xy plane, which we call z symmetry. We define the global coordinate system with axes x, y, z fixed relative to the nanobeam, where the x axis is parallel to the nanobeam. The LN material coordinate system is denoted with axis X, Y, Z (Supplement 1). Given a structure that has y symmetry, as long as the material properties also remain invariant under reflection about the xz plane, the modes can be classified according to their y symmetry. For optical waves, material properties will be invariant as long as the Z is in the longitudinal direction of the beam. For mechanical waves in LN, the situation is a bit more complicated due to the anisotropic elastic properties that generally break the y symmetry, even for a symmetric structure with the Z axis aligned along the nanobeam. In the special case where the nanobeam is fabricated on Y-cut LN and is parallel to the crystal Z axis, one of the crystal mirror planes coincides with the y symmetry plane of the nanobeam geometry, and the y symmetry is restored. Under this configuration, the optical and mechanical modes can be classified according to their y-symmetry properties.

We adapt the design recipe demonstrated for silicon OMCs [11] to LN considering our etch properties and material anisotropy. We first design a mirror unit cell with a quasi-TE optical bandgap of 30 THz at X point around 194 THz and a partial mechanical bandgap from 1.48 GHz to 2.17 GHz for y-symmetric mechanical modes. The geometry of the mirror cell is shown in Fig. 1(a), with outer width $w_1 = 1247$ nm, inner width $w_2 = 919$ nm, $h_x = 330$ nm, $h_y = 859$ nm, thickness $t = 300$ nm, and lattice constant $a = 558$ nm. We use a sinusoid-shaped outer edge to realize a larger airhole. Figures 1(c) and 1(d) show the optical and mechanical band structures of the mirror cell. The corresponding bandgaps are emphasized as pink-shaded regions.

To localize optical and mechanical modes, we focus on the X-point fundamental quasi-TE optical mode and the Γ-point mechanical breathing mode. By reducing the lattice constant and the separation between the airholes, the X-point optical mode shifts up in frequency and the Γ-point breathing mode frequency is reduced, putting both modes inside their corresponding bandgaps. Using multivariable genetic optimization, defect cell geometry parameters are found to be $(a, w_2, h_x, h_y) = (450, 1092, 334, 811)$ nm by maximizing $g_0 \min(Q, Q_{\text{th}})/Q_{\text{th}}$, where g_0 is the optomechanical coupling rate, Q is the radiation-limited optical quality factor, and $Q_{\text{th}} = 10^7$ is chosen to ignore unachievably high Q [11]. The full nanobeam is generated by a smooth transition from the mirror cell geometry to the defect cell geometry. The variation of the parameters is shown in Fig. 1(f) together with the finite-element method simulated [45] fundamental TE optical mode [Fig. 1(g)] and mechanical breathing mode [Fig. 1(h)]. The LN thickness t and the outer edge w_1 are kept the same for the full nanobeam. The global and material coordinate systems are shown at the bottom left of Fig. 1(h). The inplane rotation angle ϕ is defined as the angle between crystal axis Z and global axis x for both X-cut and Y-cut LN (Supplement 1). The dashed (dotted) arrow represents the crystal Y (X) axis for X (Y)-cut LN. We obtained a radiation-limited optical quality factor $Q \sim 4 \times 10^6$ and an effective mode volume as small as $\sim 0.2(\lambda/n)^3$ for the simulated fundamental optical mode at $\omega_c/2\pi = 200$ THz. The simulated mechanical breathing mode is at $\omega_m/2\pi = 2.1$ GHz with effective mass $m_{\text{eff}} = 645$ fg and zero-point motion $x_{\text{zp}} = 2.48$ fm. The mode profiles are modified slightly by the crystal orientation. The primary effect of the orientation, however, is the variation in the optomechanical coupling due to the anisotropic nature of the photoelastic effect; this is studied in more detail in Supplement 1.

The lack of y symmetry in general induces leakage of phonons into other propagating modes [blue bands in Fig. 1(d)] of the beam by making nominally disallowed transitions possible as has been studied in previous work on silicon structures [46]. To reduce the effect of this source of mechanical dissipation, we implement a 1D phononic shield (PS) to anchor the nanobeam [11,47]. The geometry of the PS unit cell is shown in Fig. 1(b), including the top aluminum electrode layer with thickness $t_m = 100$ nm. The dimensions of this PS unit cell are chosen to be $(a, w_{1x}, w_{1y}, w_2) = (1000, 600, 650, 50)$ nm such that the mechanical band structure of the unit cell opens a full

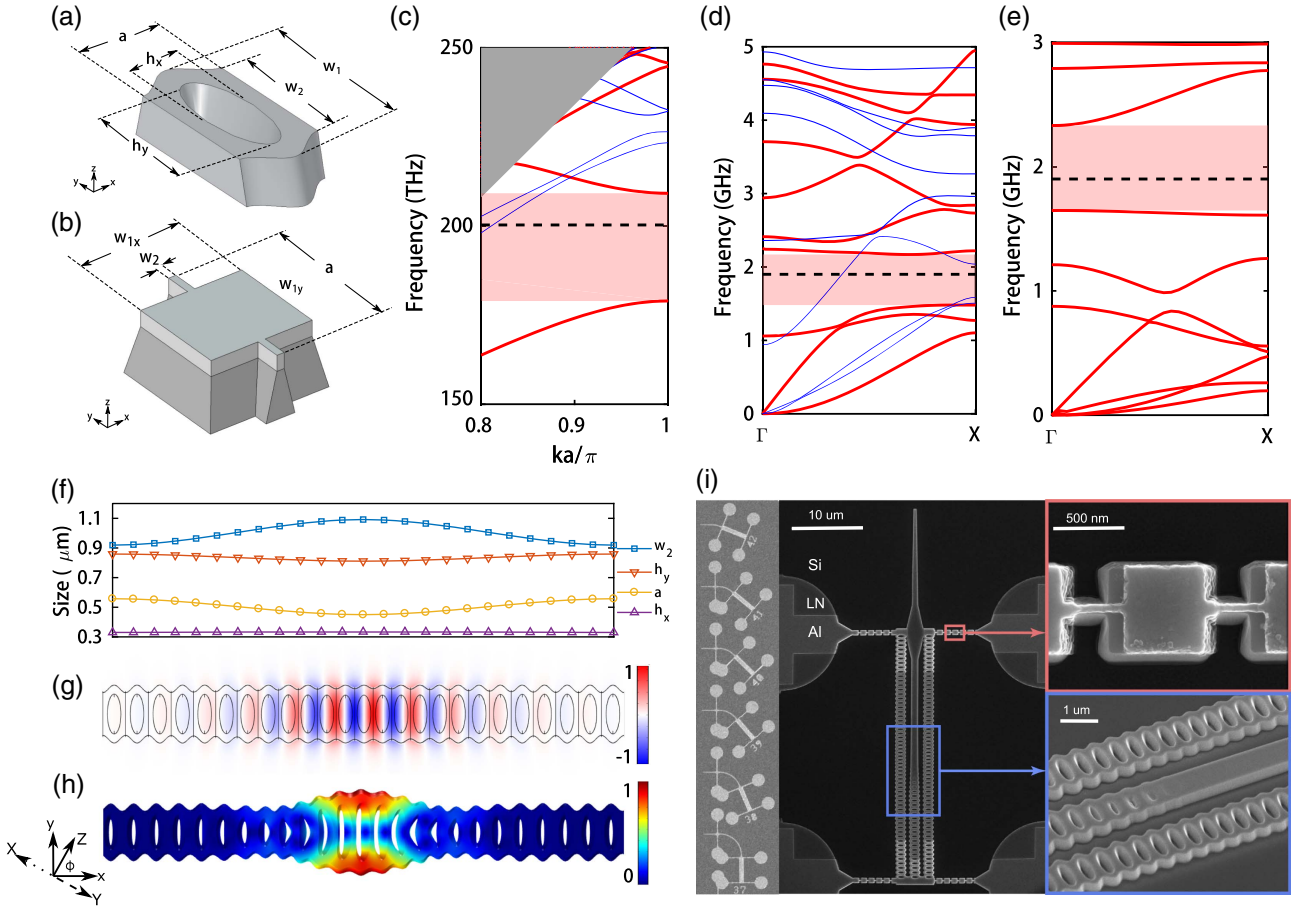


Fig. 1. One-dimensional optomechanical crystal (OMC) design. (a) Unit cell geometry of the nanobeam OMC, showing definitions of the parameters. (b) Unit cell geometry of the 1D phononic shield. The direction of the periodicity is perpendicular to the direction of the nanobeam. The (c) optical and (d) mechanical band structure of the nanobeam unit cell. The bands are classified by y symmetry (red) and y antisymmetry (blue). In (c), the gray-shaded region represents the continuum of radiation modes. The pink-shaded regions highlight the bandgap for (c) quasi-TE optical modes and (d) symmetric mechanical modes. The dashed black lines correspond to the localized fundamental TE optical mode and the mechanical breathing mode of the nanobeam, respectively. (e) Mechanical band structure of the 1D phononic shield unit cell, showing a complete bandgap around 2 GHz. (f) Variations of unit cell parameters along the nanobeam. w_1 is kept constant for all unit cells. (g) E_y component of the optical mode. (h) Displacement field of the breathing mode. The global and material coordinates are shown (see text for description). (i) Scanning electron micrograph (SEM) of the fabricated device. Left: one row of devices with different nanobeam orientations, taken before the aluminum liftoff step. Middle: top view of one device, taken before the release step. The device consists of the tapered coupler, reflector, and two nanobeams side-coupled to the reflector. The 1D phononic shields are visible between the nanobeams and the four circular anchors. Top-right: top view of the 1D phononic shield region, showing both the LN pattern and the Al electrodes. Misalignment on the order of 20 ∼ 100 nm is visible. Bottom-right: SEM of the nanobeam-reflector coupling region.

bandgap from 1.65–2.33 GHz [Fig. 1(e)]. The effect of the metallization layer has been taken into account in these simulations.

3. FABRICATION

In contrast to silicon, nanofabrication of LN is known to suffer from a lack of an effective chemical etch. However, since the development and commercial availability of thin-film LN on silicon (LNOS) oxide [48,49], high-Q microring resonators and photonic crystals have been demonstrated with physical argon milling and reactive ion etching [28,29,50–53]. We adopt an argon milling process similar to Ref. [50] for patterning the OMC structure.

The OMCs are fabricated on a LNOS wafer, with an initial LN thickness of $t \approx 500$ nm. The LN layer is first thinned to $t = 300$ nm with argon ion milling. The nanobeam geometry is defined by electron beam lithography (EBL) with HSQ resist. The pattern is then transferred to the LN layer by argon ion milling. The remaining HSQ is removed with 2 min 5:1 buffered oxide etch (BOE). A second aligned EBL with CSAR resist followed by a liftoff process defines the aluminum metal layer. The device is finally released with a masked XeF_2 dry etch that selectively removes the silicon. Electrical contacts between on-chip electrodes and aluminum wires are made with ultrasonic wirebonding. Tapered optical mode converters are patterned on the LN layer with efficiency $\eta \sim 50\%$ for coupling the OMCs to a lensed fiber [9].

Figure 1(i) shows scanning electron micrographs of the resulting structures taken before the release step. A notable feature of the sample is that it contains many devices with the identical OMC design, but where we have rotated the orientation of the nanobeam with respect to the crystal axis (see also Ref. [20] for similar measurements on GaAs). This is done to give us a better understanding of the anisotropic photoelastic property of LN and its effect on the performance of the OMCs.

4. OPTOMECHANICAL CHARACTERIZATION AND MECHANICAL LASING

We measured the devices at both room temperature and at low temperature, by cooling to 4 K inside a closed-cycle Montana Instruments cryostat. The actual temperature of the phonon mode in the cryostat is closer to ∼20 K. This is determined by calibrated thermal spectroscopy as explained in Ref. [9]. The measurement setup, shown in Fig. 2(a), is suited for understanding the optical and mechanical properties of the system. The optical properties were first characterized by scanning a continuous-wave tunable laser and monitoring the reflection spectrum. Figure 2(b) shows a typical optical reflection spectrum with one sharp resonance from the localized optical mode. A narrow scan near the resonance is fit to a Lorentzian lineshape to determine the linewidth and quality factor [Fig. 2(b) inset]. For the device under consideration, a loaded Q of 2.5×10^5 and an intrinsic Q of 3.5×10^5 are extracted, corresponding to a total linewidth $\kappa/2\pi = 776$ MHz that is significantly smaller than the mechanical frequency and puts us in the resolved sideband regime of cavity optomechanics [32]. Similar measurements taken over multiple devices, as detailed in Supplement 1, show no apparent correlation between the optical quality factors and the crystal orientations.

A well-known effect in optical resonators is the shift in their frequency proportional to intracavity photon number n_c. This nonlinearity can result from the thermo-optic, or Kerr effect. In our measurements, we observe a strikingly different relation between intracavity photon number and cavity resonance shift. As n_c is increased, we see that the cavity frequency shifts by a frequency proportional to n_c^2. The shift becomes easily measurable at roughly $n_c \gtrsim 10^4$, when it exceeds the linewidth. At these values and larger, it is observed to be quadratic. In addition to measuring the frequency shift, we use a sideband probe to measure the cavity linewidth at different optical powers and detunings (see Supplement 1 for full dataset). Minor and reversible linewidth broadening is observed (∼15% at $n_c = 1.5 \times 10^5$). We attribute the quadratic thermal shift to absorption of the intracavity photons, where this absorption rate itself has a linear dependence on n_c. This dependence distinguishes it from green-induced infrared absorption (GRIIRA) [54], in which case the absorption rate of intracavity photons would scale as n_c^2 (the second-harmonic intensity) instead of the observed n_c. Moreover, it is likely not due to absorption of second-harmonic generated photons, which would produce the correct n_c^2 thermal shift but would also require a significant reduction in the cavity linewidth to explain the observed thermal shift. A detailed understanding of the physical mechanism behind this heating and the development of methods for its mitigation (such as surface treatment [22,23] or Mg-doping [54]) will be the subject of future investigations.

The optomechanical coupling between the localized optical mode and mechanical mode gives rise to an interaction Hamiltonian $\hat{H}_{\text{int}} = \hbar g_0 \hat{a}^\dagger \hat{a}(\hat{b} + \hat{b}^\dagger)$, where g_0 is the zero-point optomechanical coupling rate, and $\hat{a}$ and $\hat{b}$ are the annihilation operator for the optical and mechanical modes, respectively. When the optical mode is driven by a laser beam up to an intracavity photon number $n_c \gg 1$, the interaction can be linearized to $\hat{H}_{\text{int}} = \hbar G(\hat{a} + \hat{a}^\dagger)(\hat{b} + \hat{b}^\dagger)$, and $G = g_0\sqrt{n_c}$ is the effective optomechanical coupling rate [32]. The optomechanical backaction leads to a frequency shift and an extra damping rate of the mechanical mode given by [32]

$$\gamma_{\text{OM}} = G^2\left(\frac{\kappa}{\kappa^2/4 + (\Delta - \omega_m)^2} - \frac{\kappa}{\kappa^2/4 + (\Delta + \omega_m)^2}\right), \quad (1)$$

where κ is the total optical mode linewidth, $\Delta = \omega_c - \omega_l$ is the cavity-laser detuning, and ω_m is the mechanical mode frequency. In the sideband-resolved regime where $\omega_m > \kappa$, only the first term is appreciable for a red-detuned pump laser with $\Delta \sim \omega_m$, while for blue side pumping, the second term dominates. As shown in Fig. 2(a), the mechanical properties of the system can be characterized by either measuring the transduced mechanical thermal motion in the optical noise power spectral density (thermal spectroscopy) or by measuring the effect of the mechanical response on a weak optical probe generated by the electro-optic modulator (coherent spectroscopy).

The optomechanical coupling rates are observed to peak at $\phi = 135°$ on X-cut LN. This is in agreement with simulations outlined in detail in Supplement 1. We focus on one device with this specific orientation and characterize its mechanical and

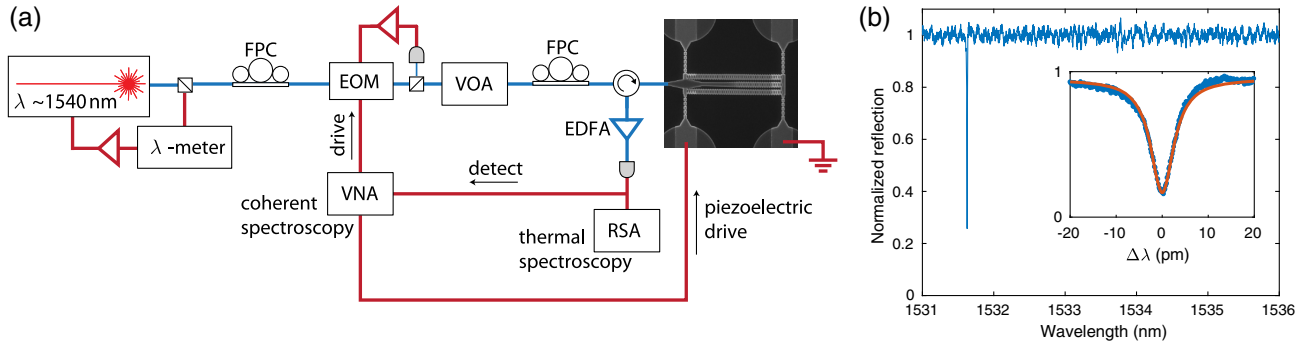


Fig. 2. Measurement setup and optical mode characterization. (a) Simplified diagram of the measurement setup. Two LN optomechanical crystals are side-coupled to a reflector. The reflection spectrum is recorded for optical characterization. The thermal mechanical motion of the nanobeam encoded in the optical noise power spectrum is measured by the realtime spectrum analyzer (RSA). A weak optical side band is generated with the vector network analyzer (VNA) and electro-optic modulator (EOM) for coherent spectroscopy. The mechanical motion can be also piezoelectrically driven, and the transduced optical sidebands are measured by the high-speed photodetector and the VNA. (b) Optical reflection spectrum of an LN OMC. A zoomed-in wavelength sweep shows a loaded optical quality factor $Q = 2.5 \times 10^5$ and corresponding intrinsic quality factor $Q_i = 3.5 \times 10^5$ (inset: blue, data; red, fit). Variable optical attenuator (VOA), erbium-doped fiber amplifier (EDFA), fiber polarization controller (FPC).

optomechanical properties. We first perform the thermal spectroscopy at low temperature with a blue-detuned laser pump, where the optomechanical backaction is simplified to

$$\gamma_{\mathrm{OM}} = -\frac{G^2\kappa}{\kappa^2/4 + (\Delta + \omega_{\mathrm{m}})^2}. \tag{2}$$

We fix the pump power $P_{\mathrm{in}} \sim 0.2$ mW and step the detuning toward $\Delta = -\omega_{\mathrm{m}}$ to increase the optomechanical cooperativity $C \equiv |\gamma_{\mathrm{OM}}/\gamma_{\mathrm{i}}|$, where γ_{i} is the intrinsic mechanical linewidth. Figure 3(a) shows three different power spectral densities of the reflected pump laser at different values of cooperativity. When $C > 1$ is reached, the total mechanical linewidth $\gamma_{\mathrm{tot}} = \gamma_{\mathrm{i}} + \gamma_{\mathrm{OM}}$ becomes negative. The system enters an oscillating regime and a mechanical lasing peak emerges in the power spectrum.

To extract the intrinsic mechanical linewidth γ_{i} and zero-point optomechanical coupling rate g_0 of the same device, we fit γ_{tot} from the power spectrum. We fix cavity-laser detuning $\Delta = \pm\omega_{\mathrm{m}}$ so the optomechanical backaction is given by $\gamma_{\mathrm{OM}} = \pm 4G^2/\kappa = \pm 4g_0^2 n_{\mathrm{c}}/\kappa$, resulting in a linear relationship between γ_{OM} and n_{c}. We adopted small pump powers $P_{\mathrm{in}} \sim 10$ μW to access stable red-detuned measurements in the presence of a thermal cavity red shift. As shown in Fig. 3(b), we measured the device at both room and low temperature and on the blue and red sides. The intrinsic mechanical loss γ_{i} is reduced by a factor of 5 after cooling the device. We observe that γ_{tot} deviates from simple predictions only considering optomechanical backaction, showing an asymmetry with respect to the intrinsic linewidth between blue- and red-detuned laser pump frequency. The deviation becomes larger when the device is cooled down, where no linewidth reduction is observed with a blue-detuned pump laser. We attribute the deviation to thermally induced linewidth broadening, which scales linearly with respect to n_{c} for low pump powers and is independent of the sign of the detuning. We note that the thermally induced broadening is more prominent in the devices with the phononic shield, which we attribute to their lower thermal conductivity, leading to a larger temperature change per photon, and their reduced smaller nominal linewidth. At higher pump powers, the thermally induced broadening saturates near $\gamma_{\mathrm{tot}}/2\pi \approx 300$ kHz and as a result, the optomechanical backaction eventually dominates the linewidth change, leading to mechanical lasing. We combine the measurements from $\Delta = \pm\omega_{\mathrm{m}}$ at low powers to eliminate the thermal broadening. We measure $g_0/2\pi = 128$ kHz, intrinsic mechanical linewidth $\gamma_{\mathrm{i}}/2\pi = 513$ kHz at room temperature, and $g_0/2\pi = 92$ kHz, $\gamma_{\mathrm{i}}/2\pi = 109$ kHz at low temperature. The cause of the temperature dependence of g_0 may be attributed to a temperature dependence of the photoelastic coefficients but is not well understood at this time. The corresponding mechanical quality factor is $Q_{\mathrm{m}} \sim 3500$ at room temperature and $Q_{\mathrm{m}} \sim 17000$ at low temperature. A device with identical design but without a 1D phononic shield is also measured at low temperature for comparison (Δ), showing $g_0/2\pi = 108$ kHz and $\gamma_{\mathrm{i}}/2\pi = 284$ kHz.

The optomechanical properties of the same device are also measured at low temperature using the optical sideband probe measurement (coherent spectroscopy). A typical frequency-dependent phase response of the probe is plotted in Fig. 3(c).

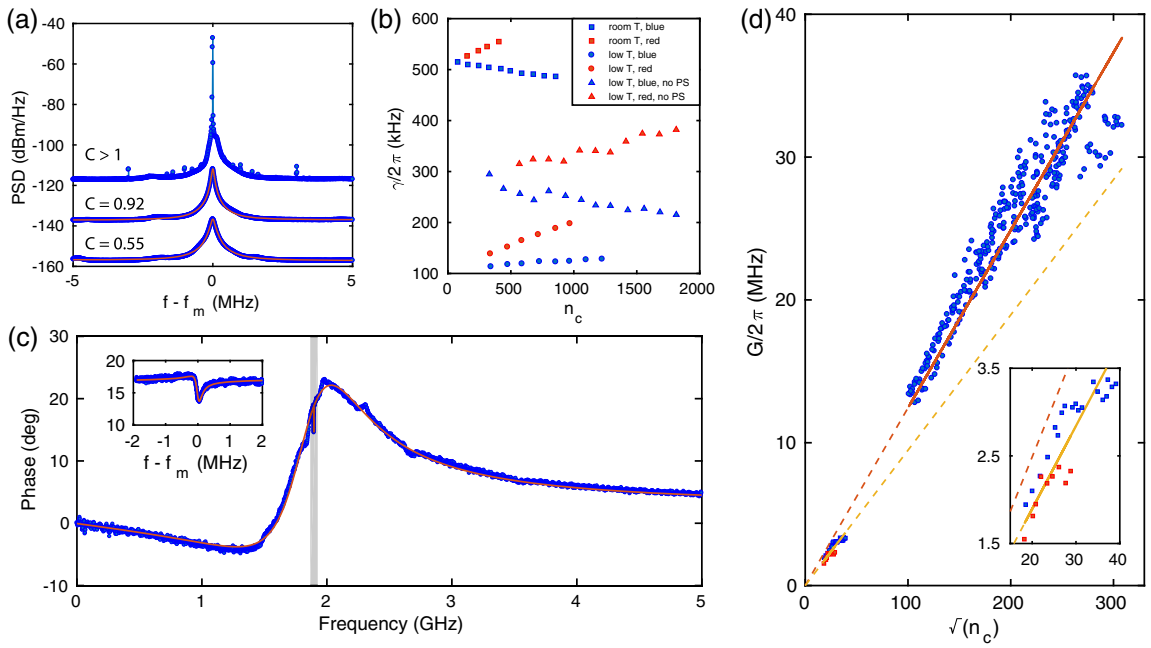


Fig. 3. EIT and optomechanical backaction measurements. (a) Power spectral density of the mechanical motion induced optical sideband. Measured spectrum with same optical power and different detunings are shown with corresponding cooperativities, vertically displaced for viewing purposes. When a pump laser beam is blue-detuned near $\Delta \approx -\omega_{\mathrm{m}}$, mechanical lasing results as the cooperativity exceeds unity. The spectrum without mechanical lasing is fitted (red) for the total mechanical linewidth γ_{tot}. (b) Total mechanical linewidth versus intracavity photon numbers under different conditions (blue, $\Delta = -\omega_{\mathrm{m}}$; red, $\Delta = \omega_{\mathrm{m}}$). For the low temperature results, one OMC without the 1D phononic shield (no PS) is measured for comparison (Δ). (c) Measured phase response of the optical probe beam at low power pump with detuning $\Delta = \omega_{\mathrm{m}}$ at low temperature (blue dots) and fit (red line). Inset: zoom-in data and fit near the transparency window. (d) Effective optomechanical coupling rate G extracted from the EIT responses at low temperature versus the square root of intracavity photon numbers $\sqrt{n_{\mathrm{c}}}$. High-power optical pump at detunings $-\omega_{\mathrm{m}} < \Delta < 0$ is adopted to achieve high n_{c} without introducing mechanical lasing. Linear fits for high n_{c} (orange) and low n_{c} (yellow) measurements are both shown for comparison. Inset: detail of the low n_{c} region, with data from both blue (blue)- and red (red)-detuned pump.

The response from the optical cavity is apparent in this measurement. There is excellent agreement between the data for the phase response for an undercoupled optical cavity ($\kappa_e < \kappa_i$). Additionally, we observe electromagnetically induced transparency, visible in the phase variation near the mechanical frequency [55–58]. The data shown was measured with optical pump detuning $\Delta = \omega_m$. The corresponding cooperativity C determines the size of the EIT feature. By fitting the phase response of the probe, we obtain an independent measurement of the optical mode linewidth κ, the intrinsic mechanical mode frequency ω_m and linewidth γ_i, and also the effective optomechanical coupling strength G [56]. The intracavity photon number n_c is calculated with κ from the EIT fit to take into account minor changes in κ.

We extracted G with different intracavity photon number n_c in Fig. 3(d). Measurements with n_c approaching 10^5 were achieved by using high pump powers at detunings $-\omega_m < \Delta < 0$ to avoid mechanical lasing. The relation between G and $\sqrt{n_c}$ is determined by fitting the low-power and high-power measurements independently. The results of these fits are shown in Fig. 3(d) as yellow and orange lines, extended for comparison. From these fits, we extracted zero-point couplings of $g_0/2\pi = 127 \pm 9$ kHz for high n_c and $g_0/2\pi = 96 \pm 10$ kHz for low n_c. We find reasonable agreement of g_0 between backaction measurement at low (high) temperature and EIT/EIA measurement at low (high) n_c. An important figure of merit is g_0/κ, which for this system is 10^{-4}, making it comparable to competing implementations in AlN, SiN_x, and GaAs [33].

5. PIEZO-OPTOMECHANICAL CHARACTERIZATION

We have fully characterized the optomechanical system by thermal and coherent spectroscopy. Beyond the conventional optomechanical measurements, the piezoelectricity of LN allows us to directly drive the mechanical motion of the OMCs, revealing more details of the mechanical spectrum. To access piezoelectricity, we pattern aluminum electrodes on the LN layer that can be driven with an oscillating voltage. The separation between the edges of the electrodes are chosen to be $d_{metal} = 22$ μm such that there is no degradation on the optical quality factor. When a voltage is applied on the aluminum electrodes in the nanobeam mirror cell region, an electric field parallel to the nanobeam is generated and deforms the optomechanical crystal. The deformation can be sensitively read out by the laser utilizing the optomechanical coupling.

The piezoelectric coupling between the OMC mechanical mode and the microwave channel (with impedance $Z_0 = 50\Omega$) is modeled as an external coupling induced decay rate γ_e. Our goal is to determine γ_e by an optomechanically calibrated measurement of the piezomechanical driving efficiency. We pump the optical mode at $\Delta \sim \omega_m$ and drive the mechanics through the microwave channel. The coherent amplitude of the mechanical motion from standard input–output theory is (Supplement 1)

$$\beta = \frac{-\sqrt{\gamma_e} c_{in}}{i(\omega_m - \omega_\mu) + \gamma_{tot}/2}, \tag{3}$$

where c_{in} is the input microwave field ($|c_{in}|^2$ is the microwave photon flux), $\omega_\mu \sim \omega_m$ is the microwave signal frequency, and $\gamma_{tot} = \gamma_i + \gamma_e + \gamma_{OM}$ is the total mechanical linewidth. The transduced optical field amplitude is

$$\alpha = \frac{-iG\beta}{i(\Delta - \omega_\mu) + \kappa/2}. \tag{4}$$

We readout the beat tone between α and the pump laser, giving us the piezo-optomechanical $S_{21}(\omega)$.

Figure 4(a) shows the piezo-optomechanical S_{21} taken on a Y-cut LN nanobeam with inplane rotation $\phi = 0$ at room temperature. We have chosen this orientation and ϕ so that the electrodes drive only the symmetric mechanical bands. The oscillating electric field not only modulates the optical mode frequency via the driven mechanical motion, but also through the electro-optic effect of LN. We repeat the same $S_{21}(\omega)$ measurement with different cavity-laser detuning Δ. The optical cavity response filters

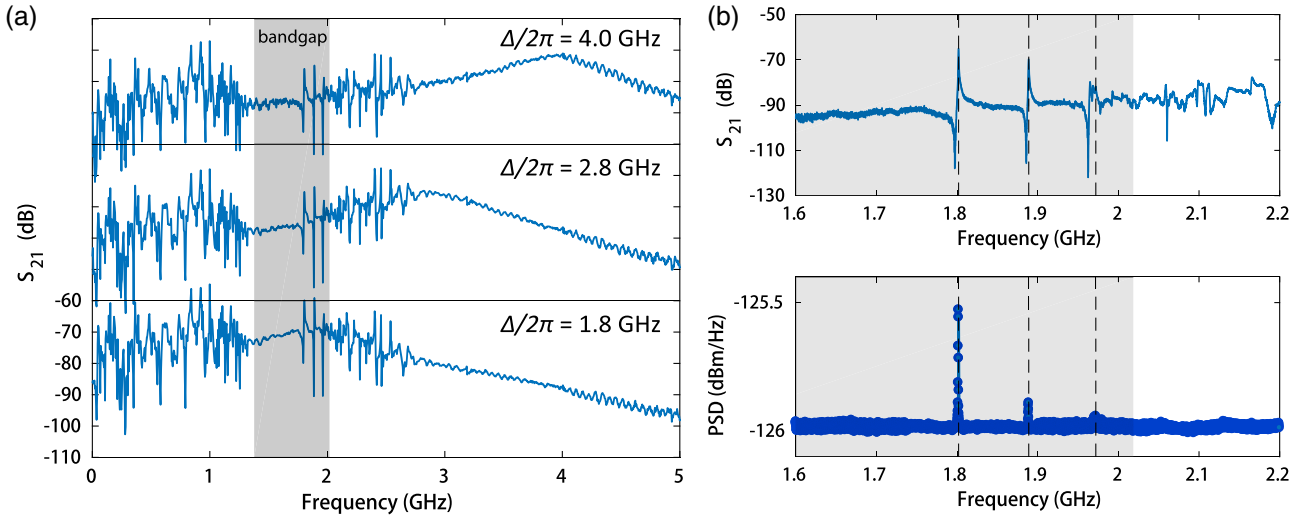


Fig. 4. Piezo-optomechanical measurements. (a) Optical readout of electro-optically and piezoelectrically driven optical sideband for different pump detunings. Broad electro-optic effect induced peak shifts when changing pump detuning, while mechanical responses stay at the same frequency. The gray-shaded area corresponds to the simulated mechanical bandgap of the nanobeam mirror unit cell. The mechanical response outside the bandgap is disordered but repeatable. (b) Piezo-optomechanical response near the upper bandgap edge. (c) Peaks of the response match well with the thermal mechanical peaks from the power spectral density, corresponding to the localized mechanical breathing modes of the nanobeam.

the electro-optically generated sidebands, leading to a broad feature with linewidth κ in the coherent response. A series of sharper peaks are also visible and arise from the piezoelectric driving of mechanical resonances of the OMC structure. We also observe a "clean" frequency range that matches reasonably well with the simulated mechanical bandgap of the nanobeam mirror cell [shaded region in Fig. 4(a)]. The localized mechanical modes can be identified near the upper edge of the bandgap. The disordered response outside the bandgap region is not due to noise of the instrument or measurement device, and is observed consistently with the same spectrum. This is likely due to complicated mixing of modes outside of the bandgap.

The mechanical origin of the narrow piezo-optomechanical response peaks can be verified by comparing the peaks to the thermal spectroscopy results. Figure 4(b) shows the piezo-optomechanical response near the upper bandgap edge, including three Fano-shaped peaks. The thermal mechanical noise power spectrum of the same frequency range is shown in Fig. 4(c). Fano-shaped response peaks in the piezo-optomechanical S_{21} match closely with the thermal–mechanical noise peaks in the power spectral density. We further calibrate the coherent phonon number from the electrical drive with the thermal phonon occupancy (Supplement 1). The decay rate $\gamma_e/2\pi = 8.8 \pm 0.56$ mHz is obtained between the microwave transmission line and the localized mechanical mode. The resulting external microwave-to-mechanics conversion efficiency is $\eta_{ext} = \gamma_e/\gamma \sim 10^{-8}$. We attribute this low efficiency to the large separation between the coupling electrodes and the impedance mismatch between the electrodes and the $Z_0 = 50\Omega$ transmission line.

If the OMC were to be connected to a microwave resonator with the same coupling electrode configuration, the microwave-mechanics coupling rate would be $g_\mu = \sqrt{\gamma_e \omega_m} \cdot \sqrt{Z_c/Z_0}/2$, where Z_c is the characteristic impedance of the microwave resonator (Supplement 1). Based on the measurement of γ_e and assuming $Z_c = 50\Omega$, we estimate $g_\mu/2\pi \sim 2$ kHz. Microwave resonators with impedance on the order of $Z_c \sim 5$ kΩ can be fabricated with high-kinetic-inductance superconducting nanowires [59] and spiral inductors [60]. By optimizing the configuration of the coupling electrodes and engineering the mechanical mode leakage [42,46], strong coupling between the microwave resonator and mechanical mode of the OMC can be realized if $\gamma_e/2\pi \gtrsim 1$ Hz and $Z_c/Z_0 \sim 100$, giving $g_\mu/2\pi \gtrsim 100$ kHz $\gtrsim \gamma/2\pi$.

6. CONCLUSIONS

In summary, we have demonstrated high-quality optomechanical crystals on a LNOS platform in the resolved-sideband regime. We measure optical quality factors above 300,000 and mechanical quality factors as high as 17,000 at cryogenic temperature. We measure optomechanical backaction, electromagnetic induced transparency, and absorption to extract $g_0/2\pi \sim 120$ kHz. We observed minor degradation ($\lesssim$15%) of the optical linewidth with intracavity photon number n_c up to 1.5×10^5. Mechanical lasing is observed with a blue-detuned laser pump, corresponding to a cooperativity above unity. Red-detuned measurements with relatively high n_c is not feasible due to the observed quadratic thermal-optic shift, which may be reduced by doping LN with MgO [54]. Given the strong nonlinear optic effect and electro-optic effect of LN, our high-quality LN OMC provides a promising platform for nonlinear photonic application and active on-chip photonic and phononic elements.

We further utilize the piezoelectricity of LN to drive the OMC. The mechanical bandgap and localized mechanical modes of the OMC are identified. Based on the measured microwave-to-mechanical conversion rate, we estimate the coupling rate between the OMC and a microwave resonator with identical coupling configuration to be $g_\mu/2\pi \sim 2$ kHz. The microwave-mechanics coupling strength can be improved by reducing the coupling electrodes separation and increasing the characteristic impedance of the microwave resonator.

With the recent advance of the strong coupling between nanomechanical structures on LN and superconducting circuits [26,38], a full microwave-to-optical conversion may be within reach with the LN OMC platform. On the optical side, the primary challenge remains heating [10,22,23,61]. Further improvements in the mechanical Q, reduced in optical absorption with improved material quality and processing, and moving toward pulsed operation may be able to address this challenge. This would open opportunities for both classical and quantum signal storing, processing, and transduction in hybrid systems exploiting densely integrated and low-energy electrical, mechanical, and optical elements [33].

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See Supplement 1 for supporting content.

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